

30 A radioactive source emits radiation of type X.

A counter is placed at a fixed position close to the source.

When a piece of paper is placed between the counter and the source, the average count rate is 24 counts per minute.

When the paper is removed, the average count rate is 532 counts per minute.

What is radiation X and the average count rate immediately after two half-lives?

	radiation X	count rate after two half-lives / min ⁻¹
A	α	127
B	β	127
C	α	151
D	β	151